

Cryptogragpy Project SEEDS

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Task 1: Frequency Analysis

When running the program the top triagram was ytn used 78 times , yt 115 and n 488 times. Second was vup 30 times . Since it's a substitution cipher.

Using this information and running the command
Command - tr 'ytnvup' 'THEAND' < ciphertext.txt > attempt1.txt

```
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ tr 'ytnvup' 'THEAND' < ciphertext.txt > attempt1.txt
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ cat attempt1.txt
THE xqaHq TzN nN qzNDqg lHmH qEEq Aguzt hNhtz ADTEH Thm IzNr qTHANgE
AlAHqz ThmE THE gArzEH DEElz lmeE A NvNAZEMhAN Txx

THE AlAHqz hAaE lJq gxxSENDed gd THE DEcmqE xB hAHfEd lEmqTEmH AT mTq xzTqET
AND THE AeeAhENT hEeLzOShN xB mHq lImc xwceANg AT THE END AND Wt Jqg qHwEd gd
THE EeChENgE xB cETxx TlceE ze gLAAzrlm exlStmHs AhcAANDg AaTtFqgE AND
A NATIOnAl cOnfENgAThAN Ag qHwEd cAd Ag A DEFEH DNEAc Aguzt lJHETHE THERE
zzzT Tz gE A eHESIDENT lImHfEd THE SEASON dIDNT cUSt SEEC EKThA lONr AT Jqg
EKThA lONr gEAUSE THE xqaHq lENE cAFEd Tz THE bHqT JEEEND mN cAHhT Tz
AFYd eHcNzAThZ lNTH THE AlxqNz eHEcOND xB THE lJNTEH xldcemaq THANqg
eDEONzrAHANr

xNE gIr JzETeAN qzHhazNDmZ Thm gEAHs AaDEcd AlAHqz mJ HxI xB mB THE
eHEcONd lmiI ADdHEq cETxx qEEaAlId AFTEH THE rOIdEN rIOgEs lHmH gEAaE
A cUgIANT xwCINrOUT eAHTd Foh TlCES Ue THE cOFECENT SWEARHEADED gd
eOWERFul hOllowOOD wCEH WHO HEIEd HAISE cIlIlONS OF DOlIAHs Tz FIrHT SEKUAl
hAHASccENT AhzND THE azxThd

qzNAlmZr THEH qzeahT rzIdEN rIngEg ATTENDEES qIATHED THEcEIfEs mN gIAAs
qUATED lAeI eINS AND SOUNDED Off AgOUT SEKlST eOWEH lcgAlANES FhOc THE hED
eAHNET AND THE STARE ON THE ARN E Jqg xAlId xT Aguzt xAg mHEJmTg ADTEH
mTg bHcEH ANAhH hATT SADIEN JUIT OnaE SHE IEARNED THAT SHE Jqg cASINr Fh
IESS THAN A cAIE aOHST AND DUHINr THE eHEcONG NATAlE eHtTcAN TOOs A gIUNT
AND SATISfDINr DIR AT THE AlIcAlE hOSTEH OF NOCINATED DINEATONS hOW aOUId
THAT gE TQeeD

Ag mT TzNqg xzT AT lEASt mN TEHQx xB THE xqaHq mT ehqAgId lONt gE

lxcEN mNfIrEd mN TncEq ze qAMD THAT AlTHxZH THE rIngEq qzNHEAdE THE
mNtRATrEs lAUNAH THEd NEFEH INTENDED IT Tz gE cUSt AN AWARDS SEASON
eACAZINr xN xNE THAT gEAaE AqzeANATED xNId lNTH hEDAHNEt AaTmNg mNTEAd
A qzEEolcAN qAMD THE zhze mJ lImhnr gEHND AlxqD DxHq AND Hqg qmHAE
AcASSED cIlIlON bHn mTg lERAl DEFENSE FUND WHIah AFTEH THE rIOgEs WAS
FlOODEd lNTH ThxzqANDq xB DnATmNg xB xH lJqg bHcE eDEIE mN qzCE
azxThmEq

Nx aAlI Tz lEAh gIAAs rXJNg lENT xzT mN ADfANaE xB THE xqaHq ThxzH THE
cOFECENT lmiI AlcOSt eHTfANId gE HEFEHNEAd gEFOHE AND DUHINr THE eHEcOND
qEHEmId qmEE FxAl cETxx qzeahTqg lmeI AqWIE ouDD lAJHA DEHN AND
NNAIE smcAN ANe qHEdZIED eHEGENTHs

ANThEH BEATZHE xB Thm gEAqN Nx xNE HEAlId smJg lIm mJ rANrZ Tz lIm gEAT
eIATuHE AhzrUgId THIS HaeEENS A lOT xB THE TlCE lNARUgId THE NAlIgITEN
NANhATIFe ONId SEHES THE AWARDS HdeE cAHShNE gUT OFTEN THE eDEIE FOhEASTINr
THE HAE SQAAlId OSaHOlOZISTS aAN cASE ONId EDUATED rJESSES

THE Jqg THE AaDEcd TAgzIATeg THE gIr lJNNEH DxqNT HEIS mN EFEND xTHEH
eATERxnd THE NvcmEE lNTH THE cAGT FxTEq lImq gZT mN THE gEAT eNAZHE
```

THE is formed in the first line.

After observation of this output trying
tr 'ytnvupxlmAqzbzH'
'THEANDOIWASF UH' < ciphertext.txt
> attempt2.txt

tr 'xHlmAqzbz' 'OHWIASFU' <
attempt1.txt > attempt2.txt

```
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ tr 'ytnvupxlmAqzbzH' 'THEANDOIWASF UH' < ciphertext.txt > attempt2.txt
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ tr 'xHlMAqzbz' 'OHWIASFU' < attempt1.txt > attempt2.txt
cat attempt2.txt
THE OSaHS TUNN ON SUNDAD WHIah SEECs AgOUT hIHT AFTEH THIS lONr STRANrE
AWAHDS ThmE THE gArzEH FEElz lIeE A NONArEMAHIAN TOO

THE AWAhDS hAaE WAS gOSeNDed gd THE DEcISE OF hAHfEd WEINSTEIN AT ITS OUTSET
AND THE AeeAhENT lceOSION OF HIS FlIC aCceAND AT THE END AND IT WAS SHAEEd gd
THE EeChENgAE OF cETOO TlCES Ue gIAAsrOWN eOIlITIAS AhcAANDg ATITIFIC AND
A NATIOnAl cOnfENgAThAN Ag qHwEd cAd Ag A DEFEH DNEAc Aguzt lJHETHE THERE
OUzHT TO gE A eHESIDENT WInfEd THE SEASON dIDNT cUSt SEEC EKThA lONr IT WAS
EKThA lONr gEAUSE THE OSaHS WEHE cOFEd TO THE FhST WESEEND IN cAHhT TO
AFYd aONFIATINr WITH THE aIOSINr eHEcOND OF THE WINTEN OldeCIaS THANAs
eDEONzrAHANr

ONE gIr JUESTION SUHhOUNDINr THIS DEAhS AaDEcd AWAhDS IS hOW Oh IF THE
eHEcOND WIlI ADdHESS cETOO ESSEaTAlId AFTEH THE rOIdEN rIOgEs WHIah gEAaE
A cUgIANT aOCINrOUT eAHTd Foh TlCES Ue THE cOFECENT SWEARHEADED gd
eOWERFul hOllowOOD wCEH WHO HEIEd HAISE cIlIlONS OF DOlIAHs Tz FIrHT SEKUAl
hAHASccENT AhzND THE aOUNThd

SIRNAIINr THEIH SUEeOhT rOIdEN rIOgEs ATTENDEES SWATHED THEcEIfES IN gIAAs
SeOhTED lAeI eINS AND SOUNDED Off AgOUT SEKlST eOWEH lcgAlANES FhOc THE hED
eAHNET AND THE STARE ON THE ARN E WAS xAlId OUT AgOUT and lEJUITO AFTEH
ITS FOhcEH ANAhH hATT SADIEN JUIT OnaE SHE IEARNED THAT SHE WAS cASINr Fh
IESS THAN A cAIE aOHST AND DUHINr THE eHEcONG NATAlE eHtTcAN TOOs A gIUNT
AND SATISfDINr DIR AT THE AlIcAlE hOSTEH OF NOCINATED DINEATONS hOW aOUId
THAT gE TQeeD

AS IT TUNNS OUT AT lEASt IN TEhcS OF THE OSaHS IT ehqAgId WONT gE

WOCEN INFOIfEd IN TlCES Ue SAID THAT AlTHOUZH THE rIOgEs SlzNIFIED THE
INITIATIFES lAUNAH THEd NEFEH INTENDED IT TO gE cUSt AN AWARDS SEASON
eACAZINr ON ONE RUAZINr ON ONE THAT eFACE ASSOCIATED ONId WITH hEDAHNEt AlTIONS INSTEAD
A SeQESMOCAN SAID THE ThOUe IS WHaINr gEHIND aIOSED DOONS AND HAS SINAE
AcASSED cIlIlON Foh ITS lERAl DEFENSE FUND WHIah AFTEH THE rIOgES WAS
FlOODEd WITH THOUSANDS OF DONATIONS OF Oh lESS FhOc eDEaIE IN SOcE
aOUNThIES

NO aAlI TO WEAh gIAAs rOWNS WENT OUT IN ADfANaE OF THE OSaHS THOUZH THE
cOFECENT WIlI AlcOSt eHTAINId gE HEFEHNEAd gEFOHE AND DUHINr THE eHEcOND
ESeeAlId SINAE FOaAl cETOO SUEeOhTENS lIeE ASHIEd ouDD lAJHA DEHN AND
NIaOIE sIDcAN ANe SAhEDUId eHESENTENS

ANOTHER FEATuRE OF THIS SEASON NO ONE HEAlId sNOWS WHO IS rOINr TO WIN gEST
eIATuHE AhzrUgId THIS HaeEENS A lOT OF THE TlCE lNARUgId THE NAlIgITEN
NANhATIFe ONId SEHES THE AWARDS HdeE cAHShNE gUT OFTEN THE eDEIE FOhEASTINr
THE HAE SQAAlId OSaHOlOZISTS aAN cASE ONId EDUATED rJESSES
```

Can see “lmth” was for with , xb for of , xNE for ONE , laq for “WAS”.

tr 'ahdcgriojewsk' 'CRYMBLGJQPZKX' < attempt2.txt > attempt5.txt closest attempt.

Few alphabets like G and L were reversed in words.

AS IT TURNS OUT AT LEAST IN TERMS OF THE OSCARS IT PROBABLY WON'T BE
WOMEN INFOGGED IN TIMES UP SAID THAT ALTHOUGH THE GOBES SIMPLIFIED THE
INITIATIVES LAUNCH THEY NEVER INTENDED IT TO BE JUST AN AWARDS SEASON
CAMPAIGN OR ONE THAT BECAME ASSOCIATED ONLY WITH REDCARPET ACTIONS INSTEAD
A SPOKESWOMAN SAID THE GROUP IS WORKING BEHIND CLOSED DOORS AND HAS SINCE
AMASSED MONEY FOR ITS GELAO DEFENSE FUND WHICH AFTER THE GOBES WAS
FLOODED WITH THOUSANDS OF DONATIONS OF OR LESS FROM PEOPLE IN SOME
COUNTRIES

NO CAGG TO WEAR BACK LOWNS WENT OUT IN ADVANCE OF THE OSCARS THOUGH THE
MOVEMENT WIGG ALMOST CERTAINLY BE REFERENCED BEFORE AND DURING THE CEREMONY
ESPECIALLY SINCE FOCAG METOO SUPPORTERS LIKE ASHLEY JUDD GAURA DERN AND
NICOLE KIDMAN ARE SCHEDULED PRESENTERS

ANOTHER FEATURE OF THIS SEASON NO ONE REALLY KNOWS WHO IS GOING TO WIN BEST
PICTURE ANYWAY THIS HAPPENS A LOT OF THE TIME IN ANYWAY THE NAUGHTY
NARRATIVE ONLY SERVES THE AWARDS HYPE MACHINE BUT OFTEN THE PEOPLE FORECASTING
THE RACE SO-CALLED OSCAR GOSSIPISTS CAN MAKE ONLY EDUCATED GUESSES

THE WAY THE ACADEMY TABOOGATES THE BILL WINNER DOESN'T HELP IN EVERY OTHER
CATEGORY THE NOMINEE WITH THE MOST VOTES WINS BUT IN THE BEST PICTURE
CATEGORY VOTERS ARE ASKED TO GIVE THEIR TOP MOVIES IN PREFERENCE ORDER IF A
MOVIE GETS MORE THAN 10 PERCENT OF THE FIRST-ROUND VOTES IT WINS WHEN NO
MOVIE MANAGES THAT THE ONE WITH THE FEWEST FIRST-ROUND VOTES IS ELIMINATED AND
ITS VOTES ARE REDISTRIBUTED TO THE MOVIES THAT LARNERED THE ELIMINATED MOVIES
SECOND-ROUND VOTES AND THIS CONTINUES UNTIL A WINNER EMERGES

IT IS A BIT TERRIBLY CONFUSING BUT APPARENTLY THE CONSENSUS FAVORITE COMES OUT
AHEAD IN THE END THIS MEANS THAT END-OF-SEASON AWARDS CHATTER INFLUENCES
INFOGGES TORTURED SPECULATION ABOUT WHICH MOVIE WOULD MOST LIKELY BE VOTERS
SECOND OR THIRD FAVORITE AND THEN EQUALLY TORTURED CONCLUSIONS ABOUT WHICH
MOVIE MIGHT PREVAIL

IN IT WAS A TOSSEUP BETWEEN BOYHOOD AND THE EVENTUAL WINNER BIRDMAN
IN WITH LOTS OF EXPERTS BETTING ON THE REFUGEE OR THE BILL SHORT THE
PRIZE WENT TO SPOTLIGHT LAST YEAR NEARLY ALL THE FORECASTERS DECARED ON
ON A HAND THE PRESUMPTIVE WINNER AND FOR TWO AND A HALF MINUTES THEY WERE
CORRECT BEFORE AN ENTIRE SNAFU WAS REVEALED AND THE RIGHTFUL WINNER
MOONLIGHT WAS CROWNED

THIS YEAR AWARDS WATCHERS ARE UNEQUALLY DIVIDED BETWEEN THREE BIGBOARDS
OUTSIDE EBBING MISSOURI THE FAVORITE AND THE SHAPE OF WATER WHICH IS
THE BALLERS PREDICTION WITH A FEW FORECASTING A HAIRY WIN FOR LET OUT

Got super close with this

tr 'GL' 'LG' <attempt5.txt> attempt6.txt

But had issues with few words as some of them weren't clear enough.

THE AWARDS RACE WAS BOOKENDED BY THE DEMISE OF HARVEY WEINSTEIN AT ITS OUTSET AND THE APPARENT IMPLOSION OF HIS FILM COMPANY AT THE END AND IT WAS SHAPED BY THE EMERGENCE OF METOO TIMES UP BLACKGOWN POLITICS ARMCANDY ACTIFISM AND A NATIONAL CONFERSATION AS BRIEF AND MAD AS A FEFER DREAM ABOUT WHETHER THERE OUGHT TO BE A PRESIDENT WINFREY THE SEASON DIDNT JUST SEEM EXTRA LONG IT WAS EXTRA LONG BECAUSE THE OSCARS WERE MOFED TO THE FIRST WEEKEND IN MARCH TO AFOD CONFLICTING WITH THE CLOSING CEREMONY OF THE WINTER OLYMPICS THANKS PYEONGCHANG

ONE BIG QUESTION SURROUNDING THIS YEARS ACADEMY AWARDS IS HOW OR IF THE CEREMONY WILL ADDRESS METOO ESPECIALLY AFTER THE GOLDEN GLOBES WHICH BECAME A JUBILANT COMINGOUT PARTY FOR TIMES UP THE MOFEMENT SPEARHEADED BY POWERFUL HOLLYWOOD WOMEN WHO HELPED RAISE MILLIONS OF DOLLARS TO FIGHT SEXUAL HARASSMENT AROUND THE COUNTRY

SIGNALING THEIR SUPPORT GOLDEN GLOBES ATTENDEES SWATHED THEMSELVES IN BLACK SPORTED LAPEL PINS AND SOUNDED OFF ABOUT SEXIST POWER IMBALANCES FROM THE RED CARPET AND THE STAGE ON THE AIR E WAS CALLED OUT ABOUT PAY INEQUITY AFTER ITS FORMER ANCHOR CATT SADLER QUIT ONCE SHE LEARNED THAT SHE WAS MAKING FAR LESS THAN A MALE COHOST AND DURING THE CEREMONY NATALIE PORTMAN TOOK A BLUNT AND SATISFYING DIG AT THE ALLMALE ROSTER OF NOMINATED DIRECTORS HOW COULD THAT BE TOPPED

AS IT TURNS OUT AT LEAST IN TERMS OF THE OSCARS IT PROBABLY WONT BE

WOMEN INFOLDED IN TIMES UP SAID THAT ALTHOUGH THE GLOBES SIGNIFIED THE INITIATIFES LAUNCH THEY NEFER INTENDED IT TO BE JUST AN AWARDS SEASON CAMPAIGN OR ONE THAT BECAME ASSOCIATED ONLY WITH REDCARPET ACTIONS INSTEAD A SPOKESWOMAN SAID THE GROUP IS WORKING BEHIND CLOSED DOORS AND HAS SINCE AMASSED MILLION FOR ITS LEGAL DEFENSE FUND WHICH AFTER THE GLOBES WAS FLOODED WITH THOUSANDS OF DONATIONS OF OR LESS FROM PEOPLE IN SOME COUNTRIES

NO CALL TO WEAR BLACK GOWNS WENT OUT IN ADfANCE OF THE OSCARS THOUGH THE MOFEMENT WILL ALMOST CERTAINLY BE REFERENCED BEFORE AND DURING THE CEREMONY ESPECIALLY SINCE FOCAL METOO SUPPORTERS LIKE ASHLEY JUDD LAURA DERN AND NICOLE KIDMAN ARE SCHEDULED PRESENTERS

ANOTHER FEATURE OF THIS SEASON NO ONE REALLY KNOWS WHO IS GOING TO WIN BEST PICTURE ARGUABLY THIS HAPPENS A LOT OF THE TIME INARGUABLY THE NAILBITER NARRATIFE ONLY SERFES THE AWARDS HYPE MACHINE BUT OFTEN THE PEOPLE FORECASTING THE RACE SOCALLED OSCAROLOGISTS CAN MAKE ONLY EDUCATED GUESSES

THE WAY THE ACADEMY TABULATES THE BIG WINNER DOESNT HELP IN EfERY OTHER CATEGORY THE NOMINEE WITH THE MOST FOTES WINS BUT IN THE BEST PICTURE CATEGORY FOTERS ARE ASKED TO LIST THEIR TOP MOFIES IN PREFERENTIAL ORDER IF A MOFIE GETS MORE THAN PERCENT OF THE FIRSTPLACE FOTES IT WINS WHEN NO MOFIE MANAGES THAT THE ONE WITH THE FEWEST FIRSTPLACE FOTES IS ELIMINATED AND ITS FOTES ARE REDISTRIBUTED TO THE MOFIES THAT GARNERED THE ELIMINATED BALLOTS SECONDPLACE FOTES AND THIS CONTINUES UNTIL A WINNER EMERGES

IT IS ALL TERRIBLY CONFUSING BUT APPARENTLY THE CONSENSUS FAVORITE COMES OUT

tr 'f' 'V' < attempt6.txt > attempt7.txt

```
seeds@seedsvm:~/Documents/Labssp-arm/files$ cat attempt7.txt
THE OSCARS TURN ON SUNDAY WHICH SEEMS ABOUT RIGHT AFTER THIS LONG STRANGE
AWARDS TRIP THE BADGER FEELS LIKE A NONAGENARIAN TOO

THE AWARDS RACE WAS BOOKENDED BY THE DEMISE OF HARVEY WEINSTEIN AT ITS OUTSET
AND THE APPARENT IMPLSION OF HIS FILM COMPANY AT THE END AND IT WAS SHAPED BY
THE EMERGENCE OF METOO TIMES UP BLACKDOWN POLITICS ARMCANDY ACTIVISM AND
A NATIONAL CONVERSATION AS BRIEF AND MAD AS A FEVER DREAM ABOUT WHETHER THERE
OUGHT TO BE A PRESIDENT WINFREY THE SEASON DIDNT JUST SEEM EXTRA LONG IT WAS
EXTRA LONG BECAUSE THE OSCARS WERE MOVED TO THE FIRST WEEKEND IN MARCH TO
AVOID CONFLICTING WITH THE CLOSING CEREMONY OF THE WINTER OLYMPICS THANKS
PYEONGCHANG

ONE BIG QUESTION SURROUNDING THIS YEARS ACADEMY AWARDS IS HOW OR IF THE
CEREMONY WILL ADDRESS METOO ESPECIALLY AFTER THE GOLDEN GLOBES WHICH BECAME
A JUBILANT COMINGOUT PARTY FOR TIMES UP THE MOVEMENT SPEARHEADED BY
POWERFUL HOLLYWOOD WOMEN WHO HELPED RAISE MILLIONS OF DOLLARS TO FIGHT SEXUAL
HARASSMENT AROUND THE COUNTRY

SIGNALING THEIR SUPPORT GOLDEN GLOBES ATTENDEES SWATHED THEMSELVES IN BLACK
SPORTED LAPEL PINS AND SOUNDED OFF ABOUT SEXIST POWER IMBALANCES FROM THE RED
CARPET AND THE STAGE ON THE AIR E WAS CALLED OUT ABOUT PAY INEQUITY AFTER
ITS FORMER ANCHOR CATT SADLER QUIT ONCE SHE LEARNED THAT SHE WAS MAKING FAR
LESS THAN A MALE COHOST AND DURING THE CEREMONY NATALIE PORTMAN TOOK A BLUNT
AND SATISFYING DIG AT THE ALLMALE ROSTER OF NOMINATED DIRECTORS HOW COULD
THAT BE TOPPED

AS IT TURNS OUT AT LEAST IN TERMS OF THE OSCARS IT PROBABLY WONT BE

WOMEN INVOLVED IN TIMES UP SAID THAT ALTHOUGH THE GLOBES SIGNIFIED THE
INITIATIVES LAUNCH THEY NEVER INTENDED IT TO BE JUST AN AWARDS SEASON
CAMPAIGN OR ONE THAT BECAME ASSOCIATED ONLY WITH REDCARPET ACTIONS INSTEAD
A SPOKESWOMAN SAID THE GROUP IS WORKING BEHIND CLOSED DOORS AND HAS SINCE
AMASSED MILLION FOR ITS LEGAL DEFENSE FUND WHICH AFTER THE GLOBES WAS
FLOODED WITH THOUSANDS OF DONATIONS OF OR LESS FROM PEOPLE IN SOME
COUNTRIES

NO CALL TO WEAR BLACK GOWNS WENT OUT IN ADVANCE OF THE OSCARS THOUGH THE
MOVEMENT WILL ALMOST CERTAINLY BE REFERENCED BEFORE AND DURING THE CEREMONY
ESPECIALLY SINCE VOCAL METOO SUPPORTERS LIKE ASHLEY JUDD LAURA DERN AND
NICOLE KIDMAN ARE SCHEDULED PRESENTERS

ANOTHER FEATURE OF THIS SEASON NO ONE REALLY KNOWS WHO IS GOING TO WIN BEST
PICTURE ARGUABLY THIS HAPPENS A LOT OF THE TIME INARGUABLY THE NAILBITER
NARRATIVE ONLY SERVES THE AWARDS HYPE MACHINE BUT OFTEN THE PEOPLE FORECASTING
THE RACE SOCALLED OSCAROLOGISTS CAN MAKE ONLY EDUCATED GUESSES

THE WAY THE ACADEMY TABULATES THE BIG WINNER DOESNT HELP IN EVERY OTHER
CATEGORY THE NOMINEE WITH THE MOST VOTES WINS BUT IN THE BEST PICTURE
```

Decrypted text file.

Task 2: Encryption using Different Ciphers and Modes

```
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ echo "I will try to encrypt this toxic message with different ciphers and modes and lets see if i can do it. " > task2.txt
```

The message I will try to encrypt using different Ciphers and modes.

To use openssl command I had to pick a key and iv value.

For the key I used the same one in the project description:

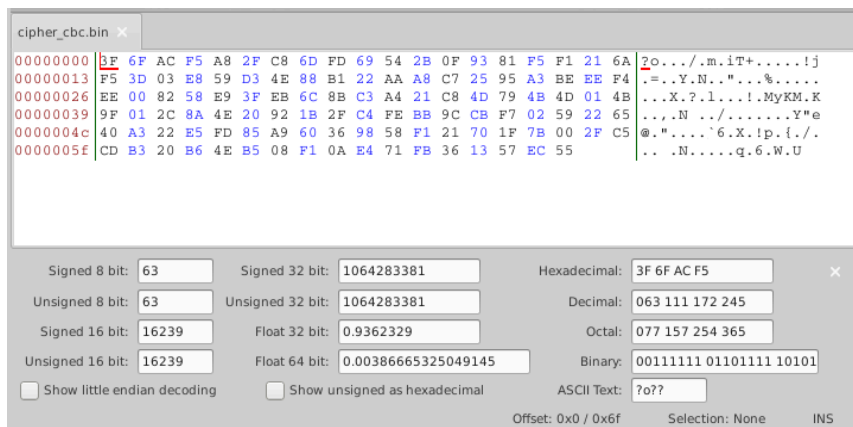
$K = 00112233445566778899aabbccddeeff$

For iv it has to be a 32 bit character then I made my own ,

$Iv = 0102030405060708090a0b0c0d0e0f10$

With AES 128 CBC

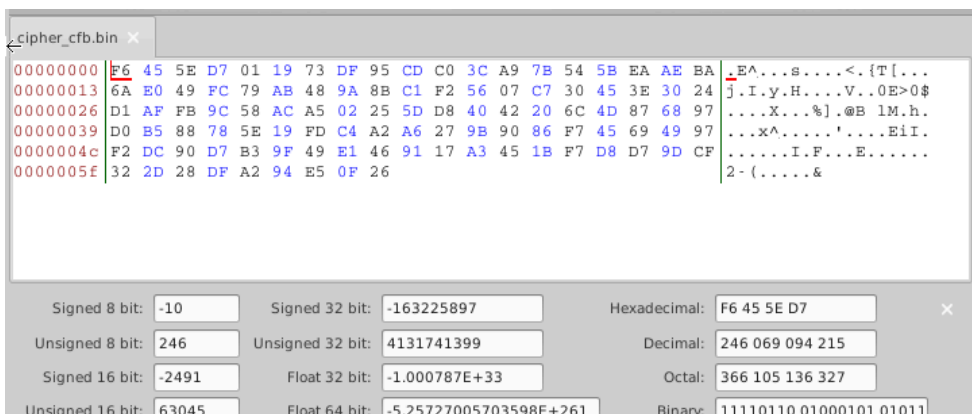
```
openssl enc -aes-128-cbc -e -in task2.txt -out cipher_cbc.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
```



With cipher block chaining mode

AES 128 CFB

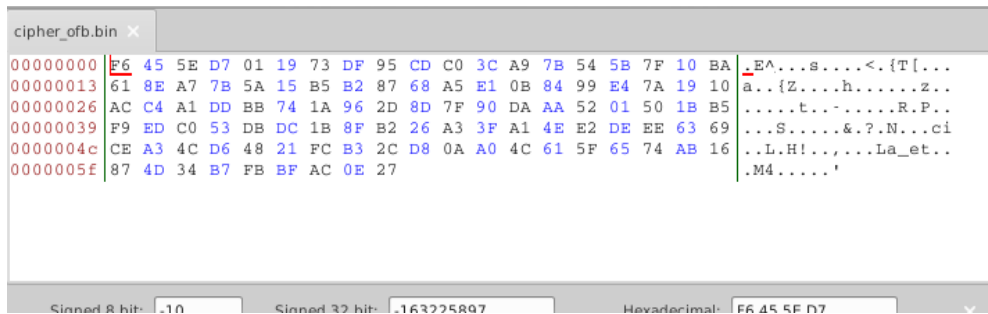
```
openssl enc -aes-128-cfb -e -in task2.txt -out cipher_cfb.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
```



The encrypted text with cipher feedback mode

AES 128 OFB

```
openssl enc -aes-128-ofb -e -in task2.txt -out cipher_ofb.bin -K
00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
```



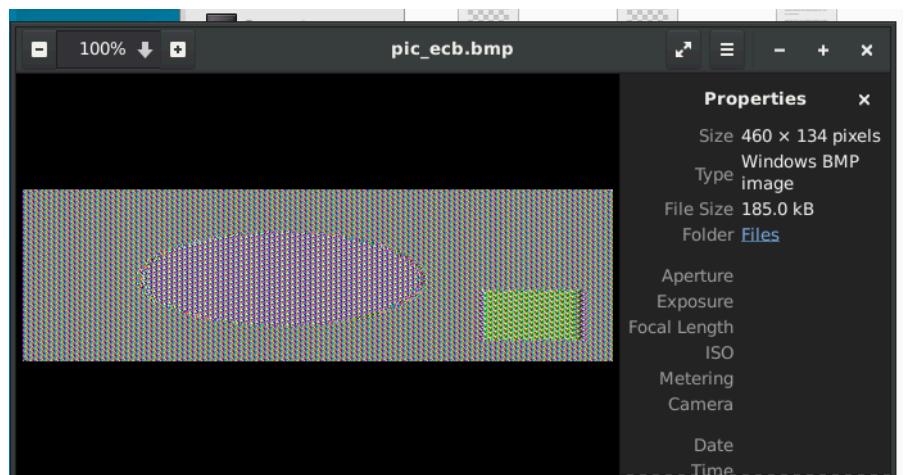
Task 3: Encryption Mode – ECB vs. CBC

Encrypting with ECB

```
openssl enc -aes-128-ecb -e -in pic_original.bmp -out pic_ecb.bin -K
00112233445566778899aabbccddeeff
```

Headers are fixed

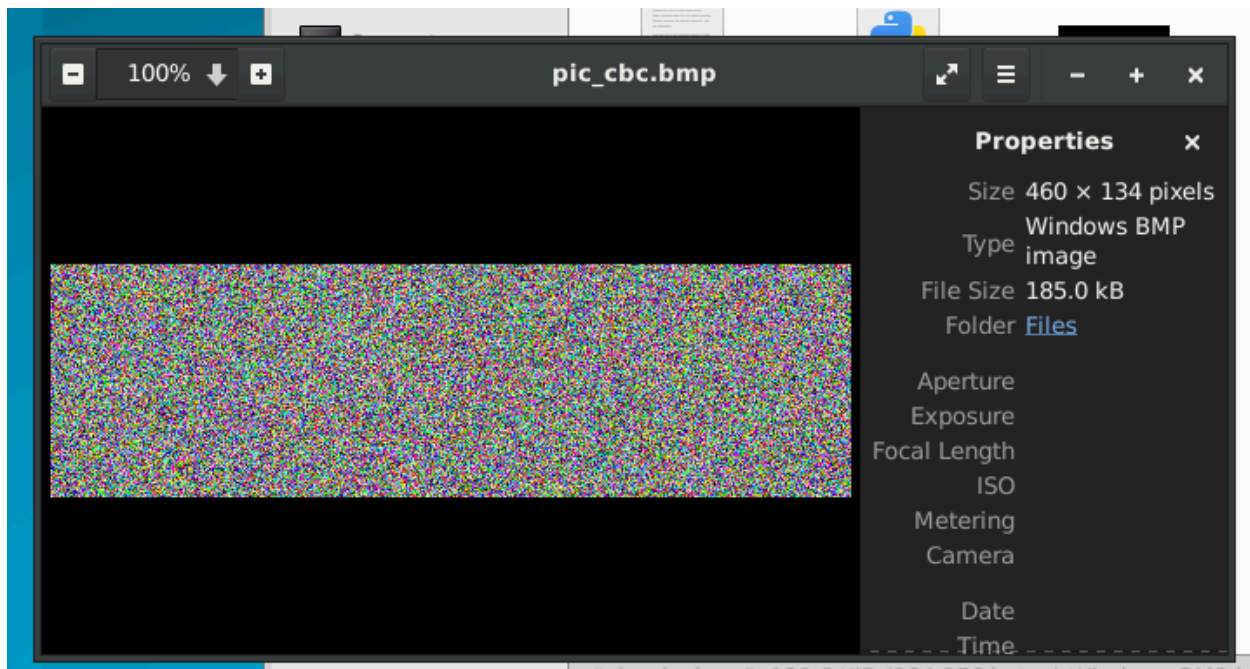
```
head -c 54 pic_original.bmp >
header
tail -c +55 pic_ecb.bin >
body_ecb
cat header body_ecb >
pic_ecb.bmp
```



ECB cannot encrypt images well
but can distinguish the colors among them from the original image and shapes are clearly visible.

Encrypting with CBC

```
openssl enc -aes-128-cbc -e -in pic_original.bmp -out pic_cbc.bin -K
00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
```



Due to chaining all blocks have become tiny pixels as each block is xor with the previous cipher so each time a different cipher block is produced which is why shapes aren't visible.

Task 4: Padding

1. Padding ECB

```
a. openssl enc -aes-128-ecb -e -in task2.txt -out t4_ecb.bin -K
00112233445566778899aabbccddeeff
```

t4_ecb.bin		
00000000	E4 FB 0A 65 85 51 87 AC C4 CB 4D 9F CA 62 45 09 A3 D2 2B	...e.Q...M..bE...+
00000013	94 40 8C FB 88 CC 6F CC C5 0C C1 41 BF 86 91 B0 4A BF 94	..@....O....A....J..
00000026	21 BB 2B CF ED B4 7B B3 BB 20 C8 4D CD F6 EA 06 EE E3 9F	!..+...{...M.....
00000039	32 91 B4 71 CC D2 A3 47 AE BD 26 82 96 CE A8 85 BC D0 DD	2..q...G..&.....
0000004c	4E 5E F4 E0 B9 40 A1 CF 08 0D 33 E9 E5 B4 FD 0B 26 15 4E	N^...@....3.....&.N
0000005f	0A 97 40 43 DF B3 0F 98 46 C7 40 70 8B B6 B8 03 FF	..@C....F.@p.....

CBC

b. `openssl enc -aes-128-cbc -e -in task2.txt -out t4_cbc.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10`

```
t4_cbc.bin x
00000000 3F 6F AC F5 A8 2F C8 6D FD 69 54 2B 0F 93 81 F5 F1 21 6A 2O.../.m.iT+....!d
00000013 F5 3D 03 E8 59 D3 4E 88 B1 22 AA A8 C7 25 95 A3 BE EE F4 .=..Y.N.."...%....
00000026 EE 00 82 58 E9 3F EB 6C 8B C3 A4 21 C8 4D 79 4B 4D 01 4B ...X.?..l...!.MyKM.f
00000039 9F 01 2C 8A 4E 20 92 1B 2F C4 FE BB 9C CB F7 02 59 22 65 ...N../.....Y"e
0000004c 40 A3 22 E5 FD 85 A9 60 36 98 58 F1 21 70 1F 7B 00 2F C5 @."....`6.X.!p.{./
0000005f CD B3 20 B6 4E B5 08 F1 0A E4 71 FB 36 13 57 EC 55 .. .N.....q.6.W.U
```

No padding

CFB Mode

`openssl enc -aes-128-cfb -e -in task2.txt -out t4_cfb.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10`

```
t4_cfb.bin x
00000000 F6 45 5E D7 01 19 73 DF 95 CD C0 3C A9 7B 54 5B EA AE BA .E^...s....<.{T[...
00000013 6A E0 49 FC 79 AB 48 9A 8B C1 F2 56 07 C7 30 45 3E 30 24 j.I.y.H....V..0E>0$
00000026 D1 AF FB 9C 58 AC A5 02 25 5D D8 40 42 20 6C 4D 87 68 97 ...X...%].@B lM.h.
00000039 D0 B5 88 78 5E 19 FD C4 A2 A6 27 9B 90 86 F7 45 69 49 97 ...x^.....'....EiI.
0000004c F2 DC 90 D7 B3 9F 49 E1 46 91 17 A3 45 1B F7 D8 D7 9D CF .....I.F...E.....
0000005f 32 2D 28 DF A2 94 E5 0F 26 2-(.....&
```

OFB Mode

`openssl enc -aes-128-ofb -e -in task2.txt -out t4_ofb.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10`

```
t4_ofb.bin x Save File
00000000 F6 45 5E D7 01 19 73 DF 95 CD C0 3C A9 7B 54 5B 7F 10 BA .E^...s....<.{T[...
00000013 61 8E A7 7B 5A 15 B5 B2 87 68 A5 E1 0B 84 99 E4 7A 19 10 a..{Z...h.....z..
00000026 AC C4 A1 DD BB 74 1A 96 2D 8D 7F 90 DA AA 52 01 50 1B B5 .....t.....R.P.
00000039 F9 ED C0 53 DB DC 1B 8F B2 26 A3 3F A1 4E E2 DE EE 63 69 ...S.....&?.N...ci
0000004c CE A3 4C D6 48 21 FC B3 2C D8 0A 0A 4C 61 5F 65 74 AB 16 ..L.H!...,...La_et..
0000005f 87 4D 34 B7 FB BF AC 0E 27 .M4.....'
```

Signed 8 bit: -10 Signed 32 bit: -163225897 Hexadecimal: F6 45 5E D7

These modes when used with AES they are used a stream ciphers and as padding is needed when using with block ciphers. They encrypt the previous cipher text to generate a key stream and then XOR with plaintext byte by byte and XOR happens with each bytes due to which blocks are not needed.

2. Three test files

```
echo -n "12345" > f1.txt      5 bytes
echo -n "1234567890" > f2.txt 10 bytes
echo -n "1234567890123456" > f3.txt 16 bytes
```

```
openssl enc -aes-128-cbc -e -in f1.txt -out f1.enc -K 00112233445566778899aabbccddeeff-iv
0102030405060708090a0b0c0d0e0f10
```

```
openssl enc -aes-128-cbc -e -in f2.txt -out f2.enc -K 00112233445566778899aabbccddeeff-iv
0102030405060708090a0b0c0d0e0f10
```

```
openssl enc -aes-128-cbc -e -in f3.txt -out f3.enc -K 00112233445566778899aabbccddeeff-iv
0102030405060708090a0b0c0d0e0f10
```

Padded bytes

```
openssl enc -aes-128-cbc -d -nopad -in f1.enc -out f1.dec -K
00112233445566778899aabbccddeeff-iv 0102030405060708090a0b0c0d0e0f10
```

```
[seeds@seedsvm:~/Documents/Labsetup-arm/Files$ hexdump -C f1.dec
00000000 31 32 33 34 35 0b 0b 0b 0b 0b 0b 0b 0b 0b 0b 0b |12345.....|
00000010
seeds@seedsvm:~/Documents/Labsetup-arm/Files$
```

```
openssl enc -aes-128-cbc -d -nopad -in f2.enc -out f2.dec -K
00112233445566778899aabbccddeeff-iv 0102030405060708090a0b0c0d0e0f10
```

```
[seeds@seedsvm:~/Documents/Labsetup-arm/Files$ hexdump -C f2.dec
00000000 31 32 33 34 35 36 37 38 39 30 06 06 06 06 06 06 |1234567890.....|
00000010
seeds@seedsvm:~/Documents/Labsetup-arm/Files$
```

```
openssl enc -aes-128-cbc -d -nopad -in f3.enc -out f3.dec -K
00112233445566778899aabbccddeeff-iv 0102030405060708090a0b0c0d0e0f10
```

```
[seeds@seedsvm:~/Documents/Labsetup-arm/Files$ hexdump -C f3.dec
00000000 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 36 |1234567890123456|
00000010 10 10 10 10 10 10 10 10 10 10 10 10 10 10 10 |.....|
00000020
seeds@seedsvm:~/Documents/Labsetup-arm/Files$
```

Task 5: Error Propagation – Corrupted Cipher Text

Before doing the task

ECB – Each cipher text block is decrypted independently and a corrupted byte effect only 1 of the 16 byte block it's inside. That entire block is garbled.

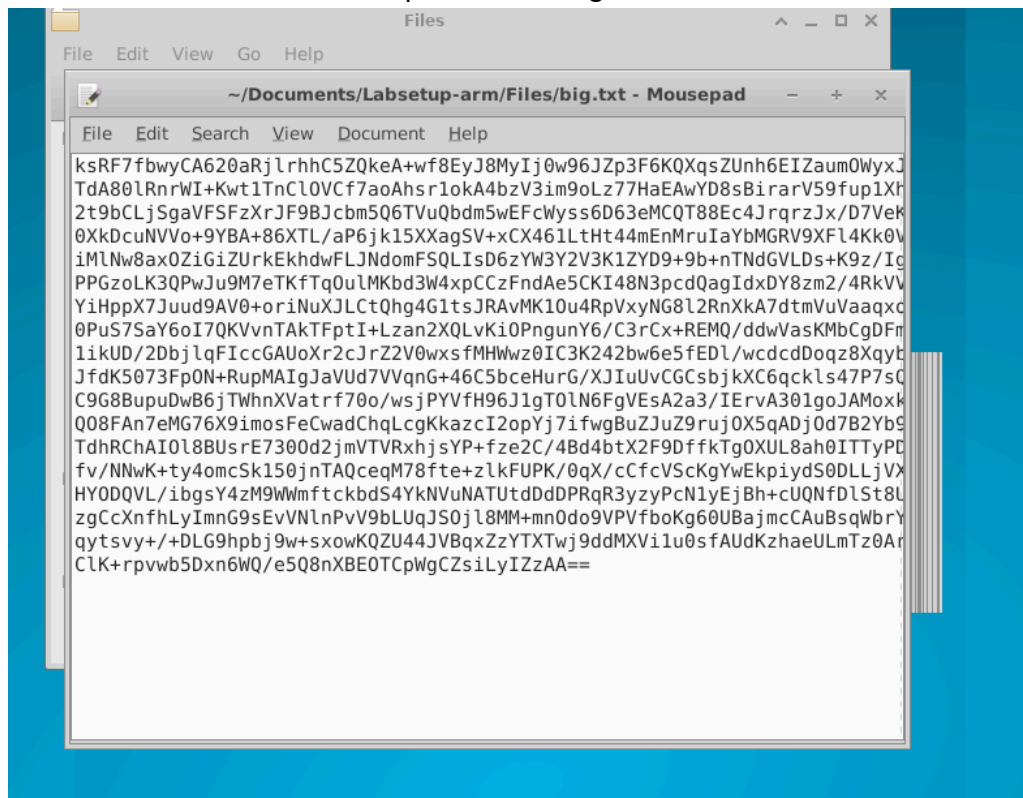
CBC – In this a corrupted block affects two blocks, the block containing the corrupted byte and block going into the next block byte. Two blocks get entirely garbled and they are adjacent to each other and both the bytes are at the same position.

CFB – I feel like this would be similar to CBC as both the blocks will be corrupted the 1st byte in the corrupted block and the block feed into the next block byte.

OFB – It doesn't depend on blocks and more on keystream, as the key stream depends on the k and iv vector. So corrupting a cipher text byte will probably just flip the same bit as the same byte location and nothing else if effected except that bit.

1. Creating a 1000 byte file

`head -c 1000 /dev/urandom | base64 > big.txt`



2. Encrpyting with ECB , CBC , CFB , OFB

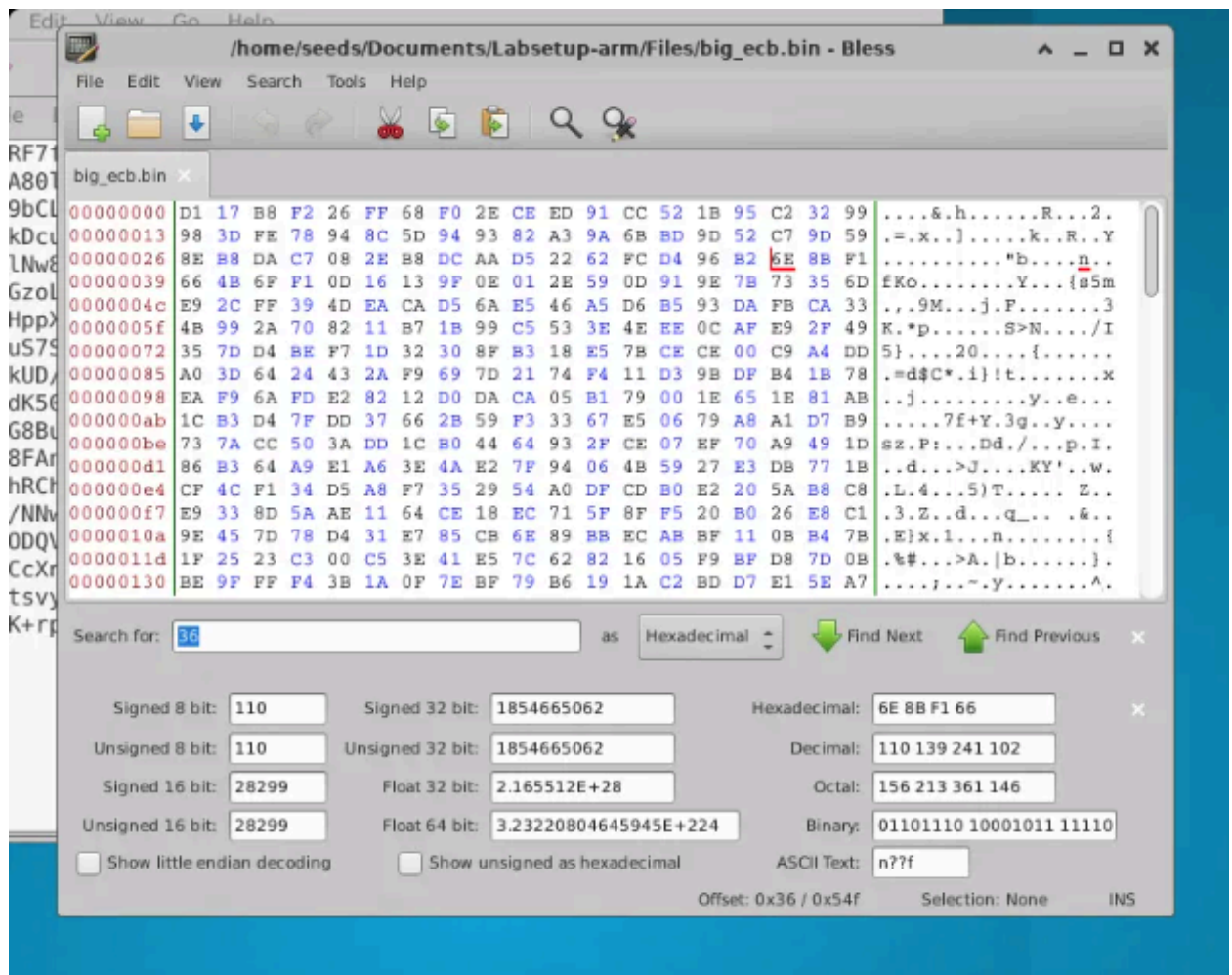
```
openssl enc -aes-128-ecb -e -in big.txt -out big_ecb.bin -K 00112233445566778899aabbccddee
```

```
openssl enc -aes-128-cbc -e -in big.txt -out big_cbc.bin -K 00112233445566778899aabbccddee  
-iv 0102030405060708090a0b0c0d0e0f10
```

```
openssl enc -aes-128-cfb -e -in big.txt -out big_cfb.bin -K 00112233445566778899aabbccddee  
-iv 0102030405060708090a0b0c0d0e0f10
```

```
openssl enc -aes-128-ofb -e -in big.txt -out big_ofb.bin -K 00112233445566778899aabbccddee  
-iv 0102030405060708090a0b0c0d0e0f10
```

3.



Changing the 55th bit in the encrypted ECB cipher file.
Similarly doing this for rest of 3 modes.

4. Now decrypting

```
openssl enc -aes-128-ecb -d -in big_ecb.bin -out dec_ecb.txt -K  
00112233445566778899aabbccddeeff
```

```
openssl enc -aes-128-cbc -d -in big_cbc.bin -out dec_cbc.txt -K  
00112233445566778899aabbccddeeff-iv 0102030405060708090a0b0c0d0e0f10
```

```
openssl enc -aes-128-cfb -d -in big_cfb.bin -out dec_cfb.txt -K  
00112233445566778899aabbccddeeff-iv 0102030405060708090a0b0c0d0e0f10
```

```
openssl enc -aes-128-ofb -d -in big_ofb.bin -out dec_ofb.txt -K  
00112233445566778899aabbccddeeff-iv 0102030405060708090a0b0c0d0e0f10
```

To understand the changes comparing the files using cmp

a. With ECB

cmp -l big.txt dec_ecb.txt

<byte position> <value in file1> <value in file2> the return values (added as a reference)

```
seeds@seedsvm:~/Documents/Labsetup-arm/Files$  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ cmp -l big.txt dec_ecb.txt  
49 106 71  
50 66 373  
51 113 106  
52 121 221  
53 130 117  
54 161 344  
55 163 143  
56 132 242  
57 125 162  
58 156 126  
59 150 21  
60 66 102  
61 105 104  
62 111 73  
63 132 167  
64 141 355  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$
```

Byte corrupted 49 ,
106 in main file , 71
hex value in the
decrypted corrupted
file

b. With CBC

cmp -l big.txt dec_cbc.txt

```
seeds@seedsvm:~/Documents/Labsetup-arm/Files$  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ cmp -l big.txt dec_cbc.txt  
49 106 331  
50 66 103  
51 113 243  
52 121 316  
53 130 75  
54 161 230  
55 163 226  
56 132 46  
57 125 145  
58 156 24  
59 150 211  
60 66 356  
61 105 362  
62 111 331  
63 132 360  
64 141 101  
71 112 115  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$
```

Entire full 16 byte wrong is
corrupted 49 – 64

c. With CFB

cmp -l big.txt dec_cfb.txt

```
seeds@seedsvm:~/Documents/Labsetup-arm/Files$  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ cmp -l big.txt dec_cfb.txt  
55 163 164  
65 165 314  
66 155 142  
67 117 132  
68 127 157  
69 171 371  
70 170 317  
71 112 45  
72 143 210  
73 105 132  
74 114 370  
75 152 76  
76 165 362  
77 12 273  
78 124 133  
79 144 127  
80 101 315  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$
```

55th byte is corrupted
and the entire next
block is changed.

d. With OFB

cmp -l big.txt dec_ofb.txt

```
seeds@seedsvm:~/Documents/Labsetup-arm/Files$  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ cmp -l big.txt dec_ofb.txt  
55 163 174  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$
```

Only 55th
byte is
effected and
changed.

Task 6: Initial Vector (IV) and Common Mistakes

Task 6.1. IV Experimented

```
seeds@seedsvm:~/Documents/Labsetup-arm/Files$  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ echo -n "I will test some IV for task 6 now" > ivtest.txt  
seeds@seedsvm:~/Documents/Labsetup-arm/Files$
```

When same IV used twice

```
openssl enc -aes-128-cbc -e -in ivtest.txt -out same_iv_1.bin -K  
00112233445566778899aabbccddeeff-iv 0102030405060708090a0b0c0d0e0f10
```

```
openssl enc -aes-128-cbc -e -in ivtest.txt -out same_iv_2.bin -K  
00112233445566778899aabbccddeeff-iv 0102030405060708090a0b0c0d0e0f10
```

Two different IV's

```
openssl enc -aes-128-cbc -e -in ivtest.txt -out diff_iv_1.bin -K 00112233445566778899aabbccddeeff -iv 0102030405060708090a0b0c0d0e0f10
```

```
openssl enc -aes-128-cbc -e -in ivtest.txt -out diff_iv_2.bin -K 00112233445566778899aabbccddeeff -iv 1112131415161718191a1b1c1d1e1f10
```

Now comparing each case

```
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ cmp same_iv_1.bin same_iv_2.bin
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ cmp diff_iv_1.bin diff_iv_2.bin
diff_iv_1.bin diff_iv_2.bin differ: byte 1, line 1
seeds@seedsvm:~/Documents/Labsetup-arm/Files$
```

When comparing the first case of same IV's , encrypting the same plaintext with the same iv produces the same ciphertext every single time as cmp operation didn't output anything.

When encrypting the same plain text and key with two different IV's different cipher text is generated and all the bytes are different.

IV needs to be unique because if an attacker can figure out the cipher text then they can get hold of the plain text and the key as the plain text are identical without even breaking the cipher text.

Task 6.2. Common Mistake: Use the Same IV

Yes reusing the IV breaks the confidentiality. As in OFB mode key stream is generated which is a function of the key and iv and it does not depend on any plain text or cipher text.

If

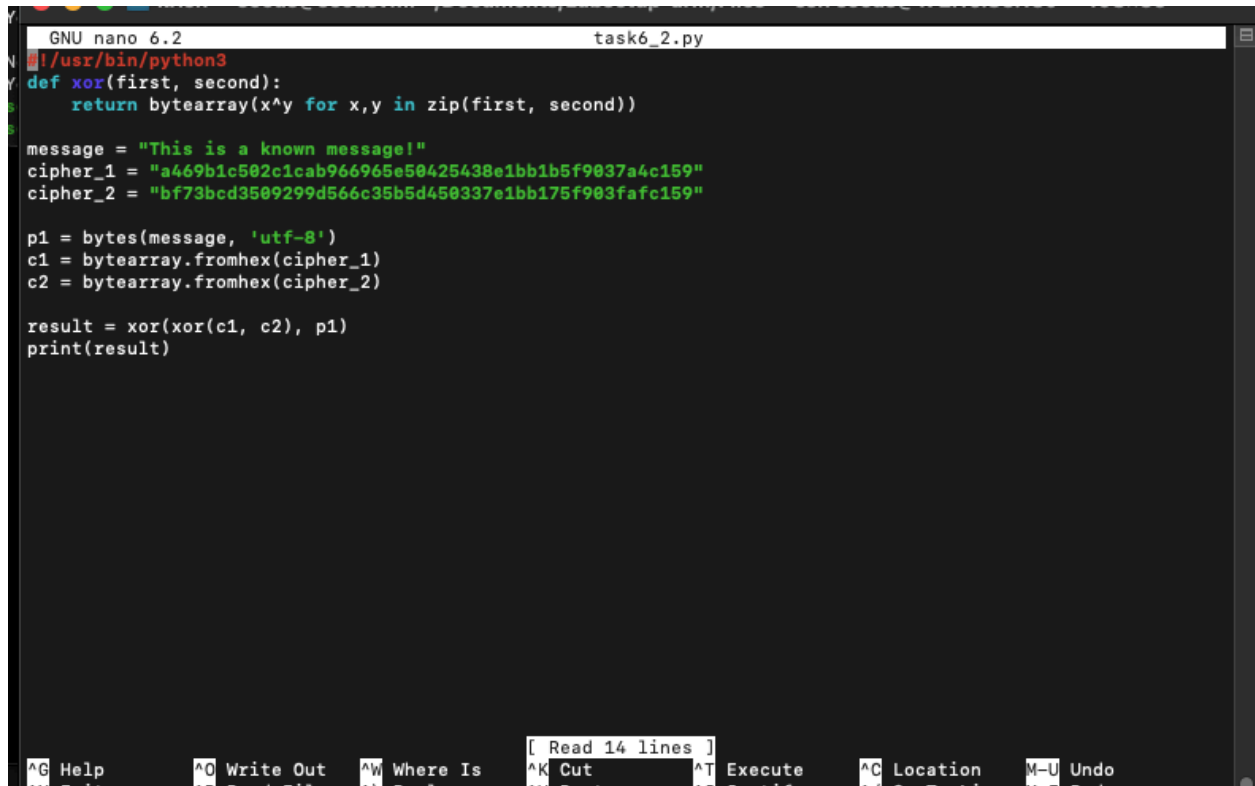
$C1 = P1 \text{ XOR KEYSTREAM}$

$C2 = P2 \text{ XOR KEYSTREAM}$

If $C1 \text{ XOR } C2$ it cancels the keystream as they are reused and the plain text is revealed.

Using the sample code with they values I ran the script

```
seeds@seedsvm:~/Documents/Labsetup-arm/Files$ python3 task6_2.py
bytearray(b'Order: Launch a missile!')
```



```
GNU nano 6.2 task6_2.py
#!/usr/bin/python3
def xor(first, second):
    return bytearray(x^y for x,y in zip(first, second))

message = "This is a known message!"
cipher_1 = "a469b1c502c1cab966965e50425438e1bb1b5f9037a4c159"
cipher_2 = "bf73bcd3509299d566c35b5d450337e1bb175f903fafc159"

p1 = bytes(message, 'utf-8')
c1 = bytearray.fromhex(cipher_1)
c2 = bytearray.fromhex(cipher_2)

result = xor(xor(c1, c2), p1)
print(result)
```

Code I ran to showcase if using the same iv and getting the message

If CFB is used instead of OFB, only the first 16 byte of p2 will be revealed as in CFB and OFB the block is generated from $C1 \oplus C2 \oplus P1$ it works on the first block but in CFB since it's chained from the next block keystream depends on the previous cipher text block . But since P1 and P2 differ and the keystream wont be canceled unlike OFB as when chained the text keeps changing every time and It won't be the same and the rest of p2 will be protected.

Task 6.3. Common Mistake: Use a Predictable IV

```
GNU nano 6.2 task6_3.py
#!/usr/bin/python3
def xor(first, second):
    return bytearray(x^y for x,y in zip(first, second))

yesorno = "5965730d0d0d0d0d0d0d0d0d0d0d0d0d"
currentiv = "40ed7da7ddfd82c9aa95224550f7b63c"
nextiv = "dbd907addfd82c9aa95224550f7b63c"

guess = bytearray.fromhex(yesorno)
current_iv = bytearray.fromhex(currentiv)
next_iv = bytearray.fromhex(nextiv)

result = xor(xor(guess, current_iv), next_iv)
print(result.hex())

[ Read 14 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
[seeds@seedsvm: ~/Documents/Labsetup-arm/encryption_oracle$ python3 task6_3.py
c25109070d0d0d0d0d0d0d0d0d0d0d0d0d
[seeds@seedsvm: ~/Documents/Labsetup-arm/encryption_oracle$ nano task6_3.py
```

Plain text to input to the encryption.

To find the Plaintext to input to the oracle encryption I used this script to generate to figure out the actual cipher text of Bob.

The message is in ascii hex 32 bytes with padding according to pkcs#5 padding. I had to pad the message as the ciphertext generating needed that bits to generate the actual message.

Cases when I hadn't added any padding neither message yes or no would work in retrieving the actual plain text message.

Also padding with some hex ascii value rather than just 0 was helpful.

```
[seeds@seedsvm: ~/Documents/Labsetup-arm/encryption_oracle$ nc 10.9.0.80 3000
Bob's secret message is either "Yes" or "No", without quotations.
Bob's ciphertext: 1e1baef735a35589aaf844cd6645950d
The IV used      : 40ed7da7ddfd82c9aa95224550f7b63c

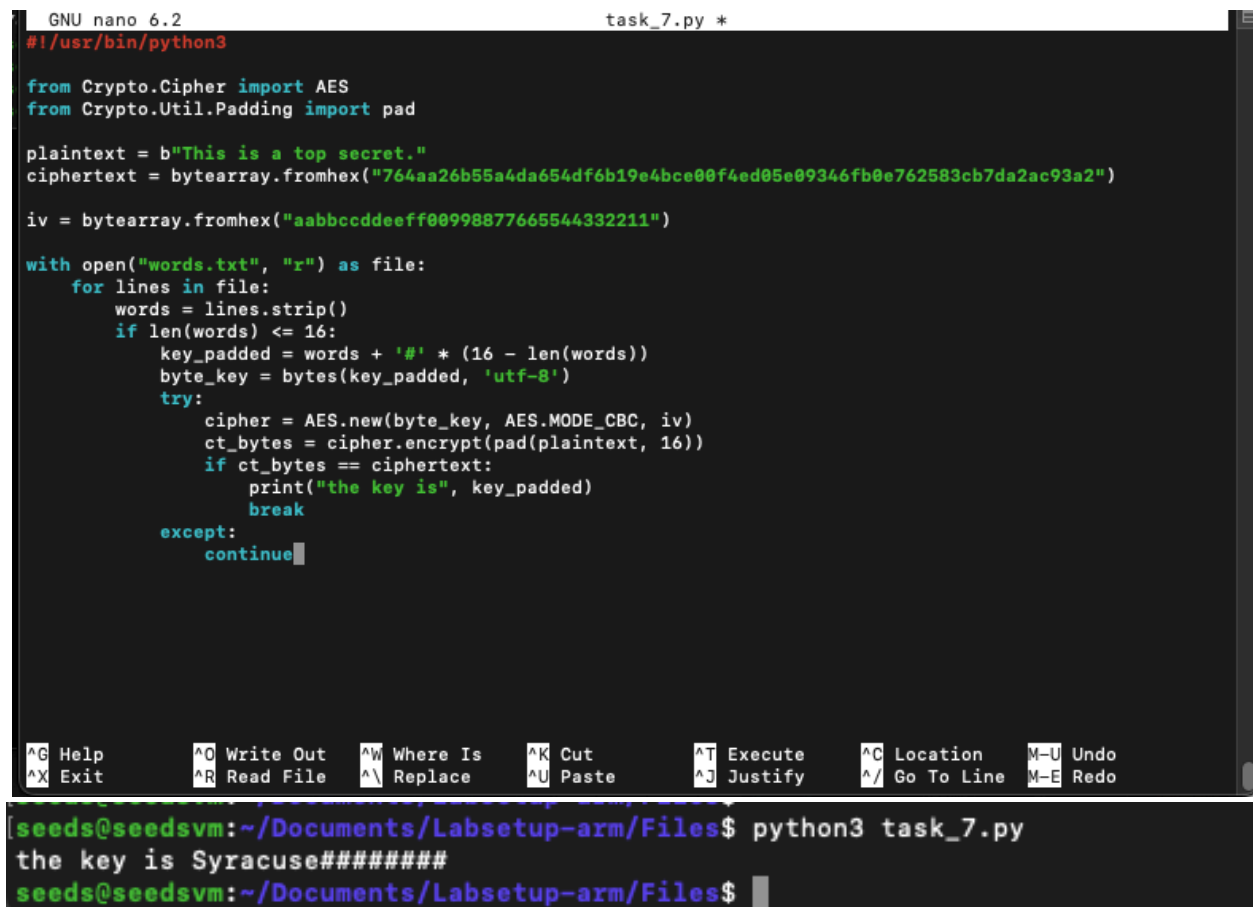
Next IV         : dbd907addfd82c9aa95224550f7b63c
Your plaintext  : c25109070d0d0d0d0d0d0d0d0d0d0d0d
Your ciphertext: 1e1baef735a35589aaf844cd6645950db67887f0960c7d9774d8cad0efc39aac
c

Next IV         : 10919fd7ddfd82c9aa95224550f7b63c
Your plaintext  : 
```

I was lucky as my first try I got the message of Bob saying “Yes” as when my cipher text first 32 bytes are compared to Bob’s cipher text the message “Yes” is revealed as I had first tried with the message “Yes”.

The Common mistake of predictable IV checks out.

Task 7: Programming using the Crypto Library



```
GNU nano 6.2 task_7.py *
#!/usr/bin/python3

from Crypto.Cipher import AES
from Crypto.Util.Padding import pad

plaintext = b"This is a top secret."
ciphertext = bytearray.fromhex("764aa26b55a4da654df6b19e4bce00f4ed05e09346fb0e762583cb7da2ac93a2")

iv = bytearray.fromhex("aabbccddeeff00998877665544332211")

with open("words.txt", "r") as file:
    for lines in file:
        words = lines.strip()
        if len(words) <= 16:
            key_padded = words + '#' * (16 - len(words))
            byte_key = bytes(key_padded, 'utf-8')
            try:
                cipher = AES.new(byte_key, AES.MODE_CBC, iv)
                ct_bytes = cipher.encrypt(pad(plaintext, 16))
                if ct_bytes == ciphertext:
                    print("the key is", key_padded)
                    break
            except:
                continue

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo

seeds@seedsvm:~/Documents/Labsetup-arm/Files$ python3 task_7.py
the key is Syracuse#####
seeds@seedsvm:~/Documents/Labsetup-arm/Files$
```

To complete this I used the python’s Pycrypto library and wrote a script.

My plain text is converted into bytes and similar to task 6.2 and 6.3 the ciphertext and iv is changed to bytes.

Since a text file called words.txt a English dictionary file it's helpful in determining the encrypted key. Using python with open function opening the text file in read mode and parsing

through all the characters in the file line by line and removing extra spaces and keeping them in an string.

Since AES cipher with cbc mode Is used for encrpyion and condition is given that the English word is shorten than 16 characters I use that condition and add padded bytes of '#' at the end of the words to form a key of 128 bits.

I use the Pycrypto's library Crypto.Cipher.AES function to make a block cipher base this functions parameters takes in key , mode CBC in our case and iv value.

Also padding is added to align with the text using Crpyto.Util.Padding. This function takes in pad bytes 16 due to AES with the plain text bits.

If this padded bytes with plaintext matches our cipher text then our key is found.

The key with this plain text and cipher text is:

Syracuse with pad of '#' 8 times to fulfill 32 bytes.

References

<https://pycryptodome.readthedocs.io/en/latest/src/cipher/classic.html>

<https://pycryptodome.readthedocs.io/en/latest/src/util/util.html>

<https://man7.org/linux/man-pages/man1/cmp.1.html>

